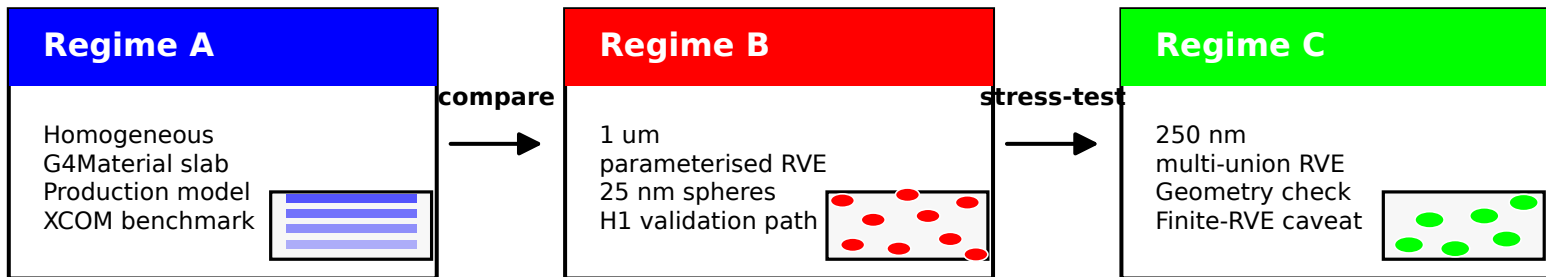


Effective-medium Geant4 predicts K-edge nanocomposite shielding

Six HDPE/high-Z fillers tested across homogeneous, parameterised-RVE, and multi-union-RVE regimes



Main decision

Use Regime A for macroscopic shielding; use Regime B for explicit validation; reserve Regime C for small-RVE checks. Multi-filler explicit RVEs must preserve composition equivalence before claiming H1 validation.

XCOM validation

6/6 pass

Mean error

0.339-0.793%

Regime B H1

5 binaries pass

Ternary case

negative control